

EE 2031 Electric Circuits

Week 1

- Introduction
- Circuit variables
- Circuit elements

Preliminaries

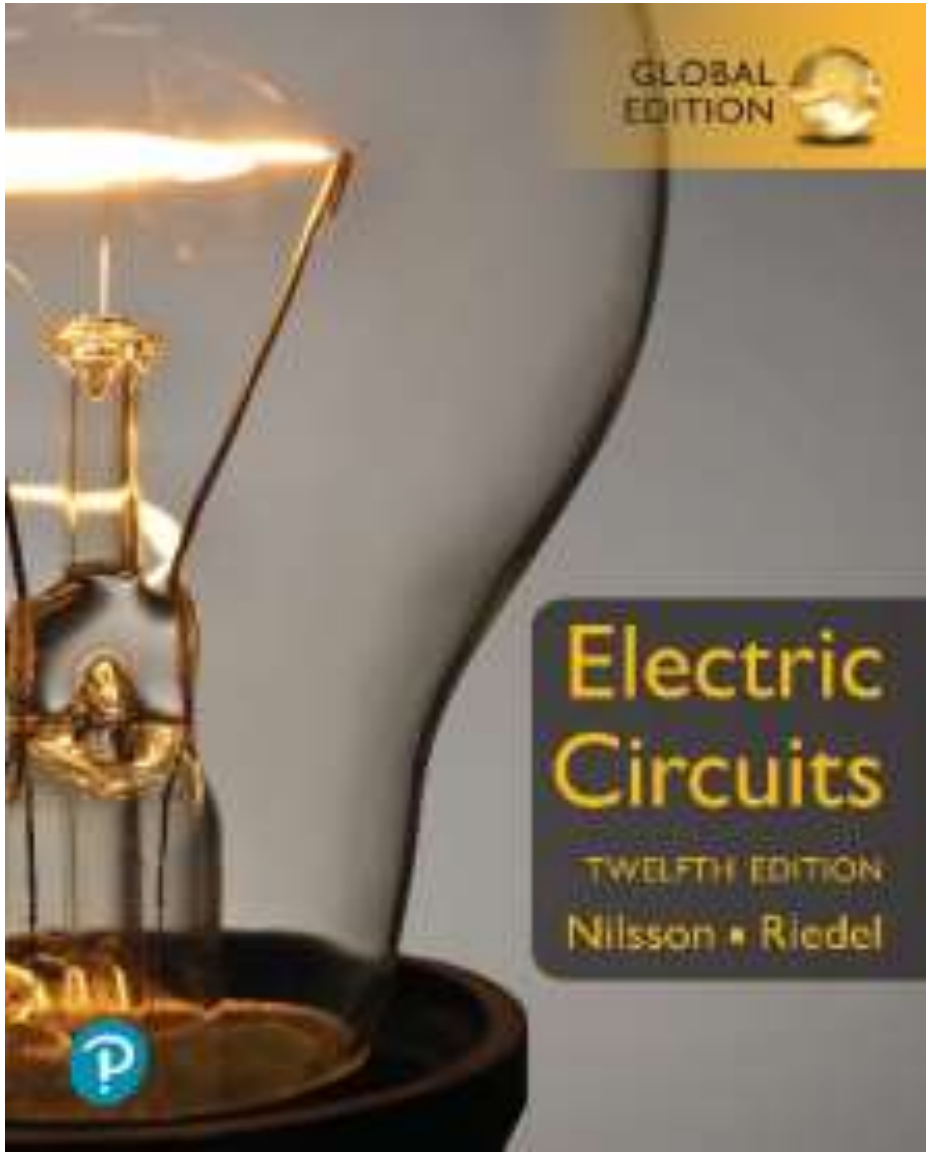
Class schedule: Monday 15:00 – 17:00 (M1.Z01)
Tuesday 15:00 – 16:00 (M2.Z11)

Instructor: Dr. Kamil Erdayandı,
kamil.erdayandi@marmara.edu.tr
Office: M2-232, Phone: (216) 777 3535

Preliminaries

Attendance:	Classroom attendance (%70) is mandatory.
Quizzes:	There will be three quizzes.
Homework(s):	There may be some homework assignments.
P.S.	There may be problem sessions before the exams.

Recommended textbook



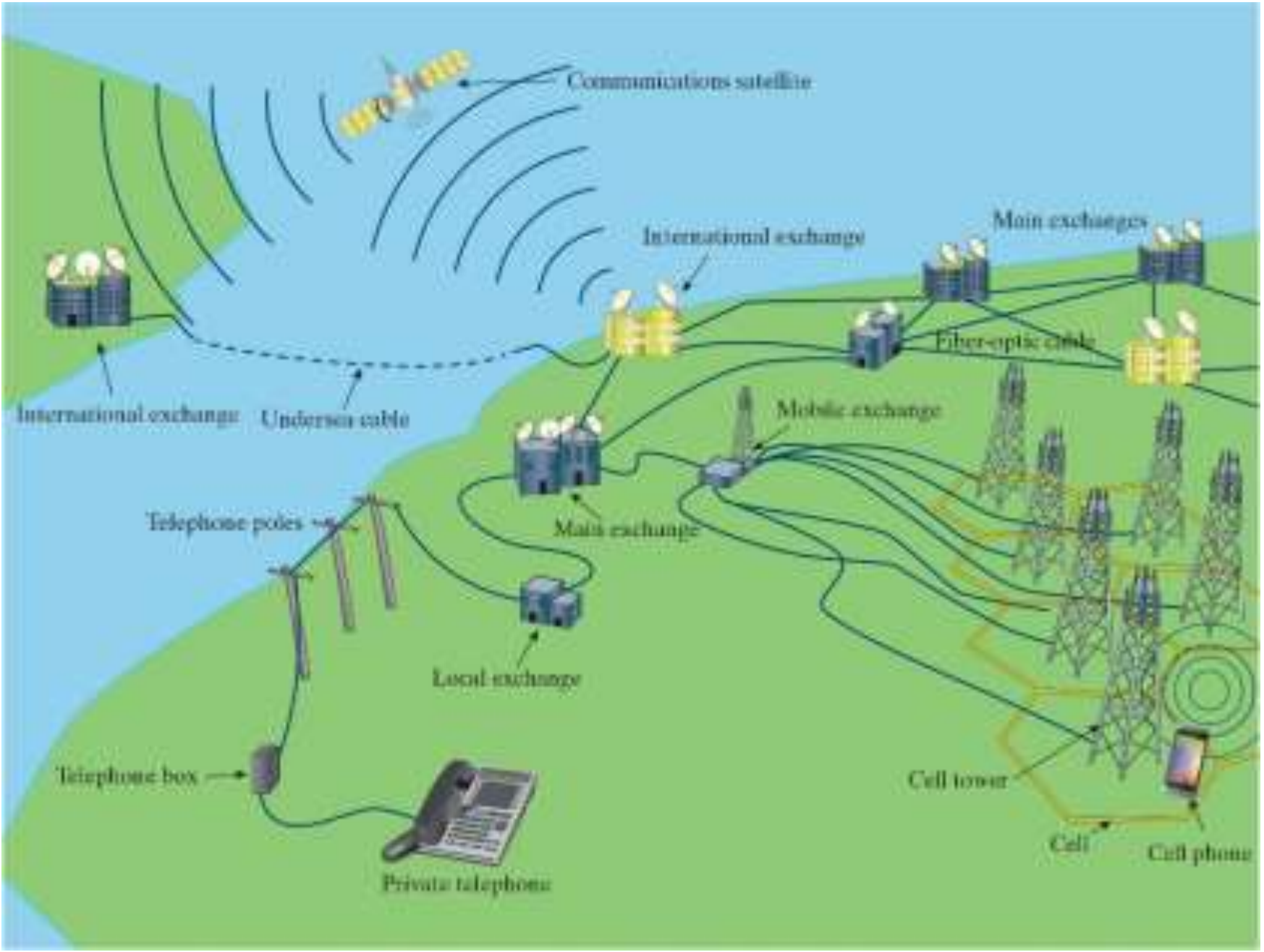
Syllabus

Grading:

Quizzes	15%
HW	(N_H x 5) %
Midterm	45 - (N_H x 5) %
Comprehensive Final	40%

N_H : Number of homework assignments.

Electric Circuits Applications – Communication Systems

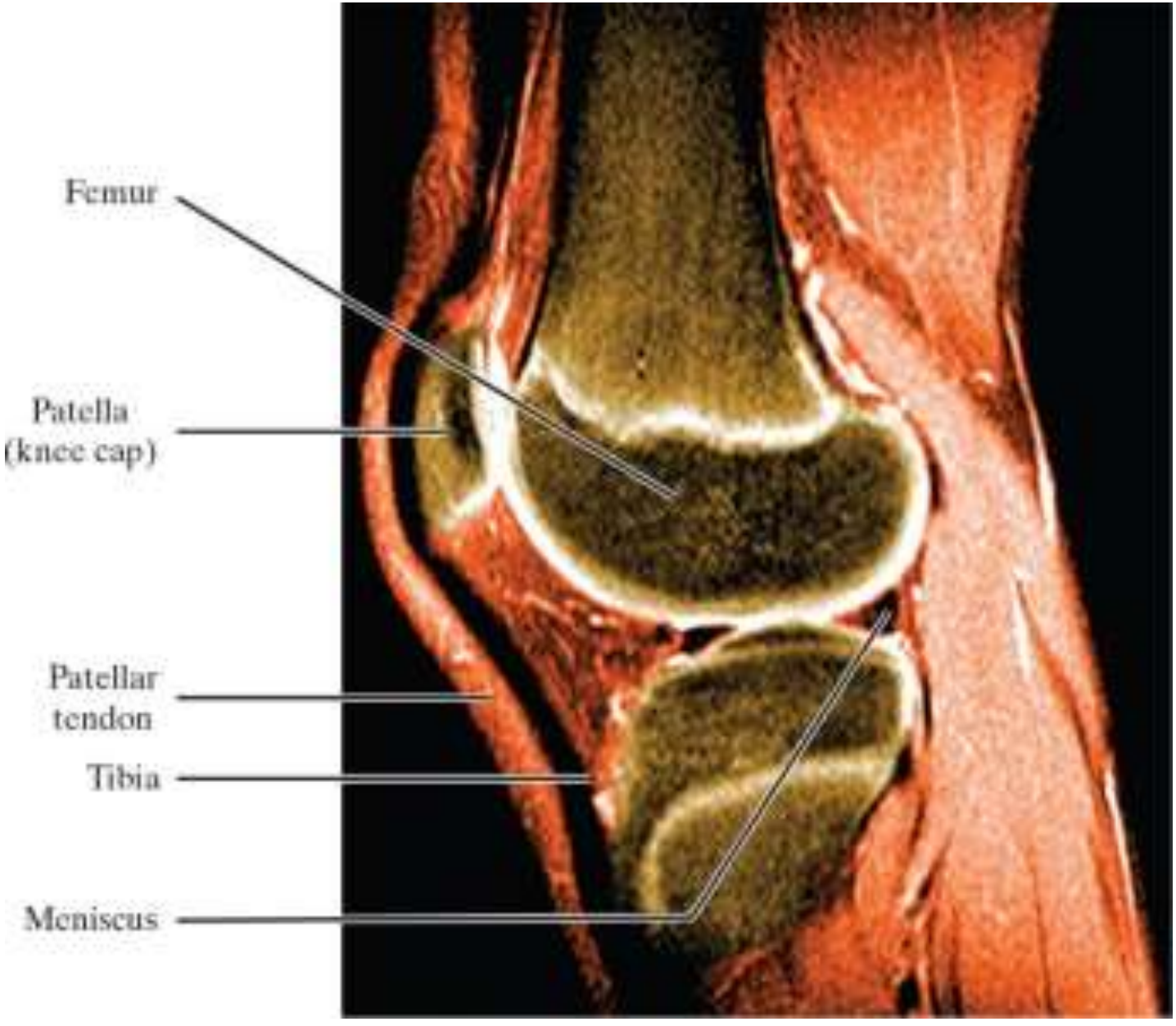


Electric Circuits Applications – Computing Systems

Our current modern PCs could be made possible thanks to electric circuit.



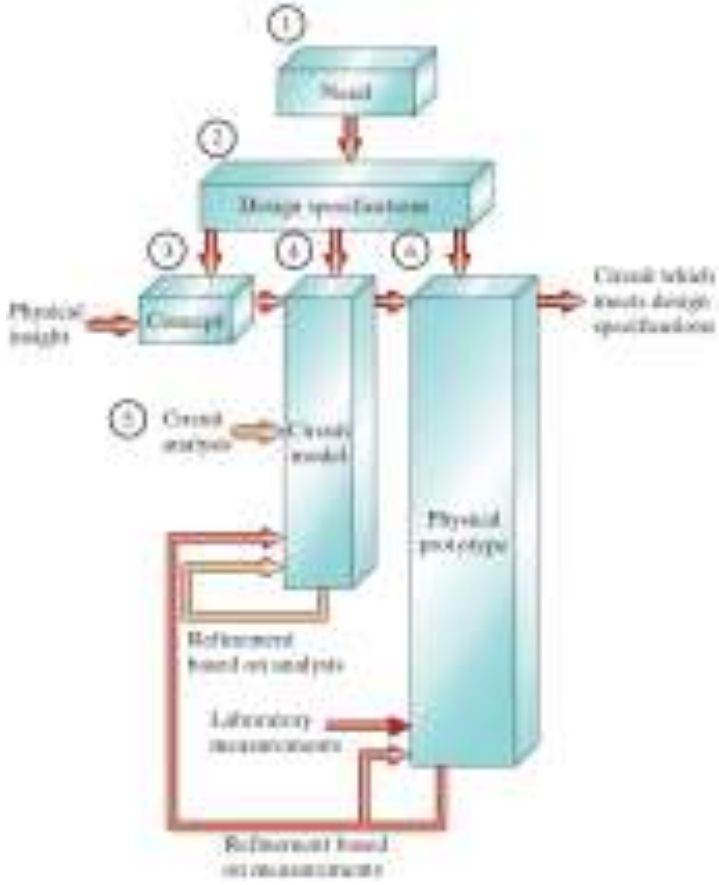
Electric Circuits Applications – Signal Processing Systems



Chapter 1 – Circuit Variables (3 of 21)

Section 1.3

How does circuit theory fit into the overall scheme of things?



Chapter 1 – Circuit Variables (2 of 21)

Memorize the boxed prefixes – we use them ALL THE TIME!!

Table 1.3 Standardized Prefixes to Signify Powers of 10

Prefix	Symbol	Power
atto	a	10^{-18}
femto	f	10^{-15}
pico	p	10^{-12}
nano	n	10^{-9}
micro	μ	10^{-6}
milli	m	10^{-3}
centi	c	10^{-2}
deci	d	10^{-1}
deka	da	10
hecto	h	10^{2}
kilo	k	10^{3}
mega	M	10^{6}
giga	G	10^{9}
tera	T	10^{12}

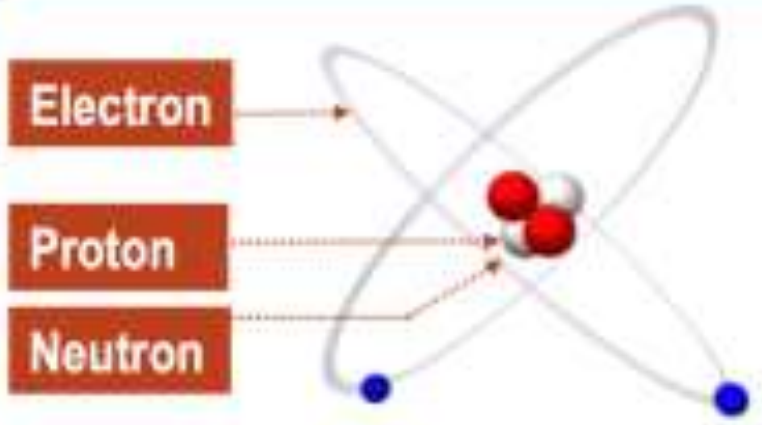
Basic Principles

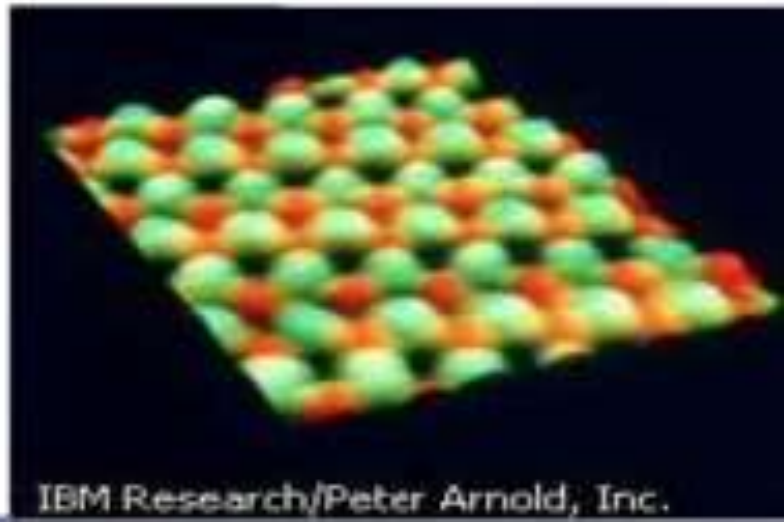
Atom

Structure of atom

Electron is assumed to be negatively charged
Proton is assumed to be positive charged

Helium Atom





Basic Principles

Electrical Charge

Definition

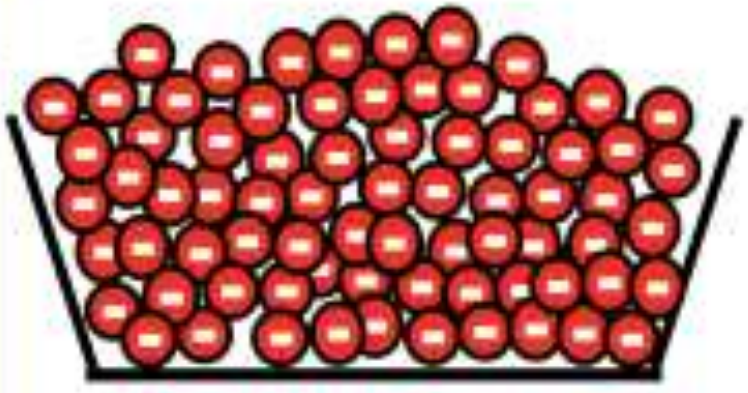
Unit of Electrical Charge Coulomb

6.3×10^{18} electrons $\overset{\Delta}{=}$ 1 Coulomb

or

Electrical charge / electron = $1/(6.3 \times 10^{18})$ Coulomb

$= 1.602 \times 10^{-19}$ Coulomb

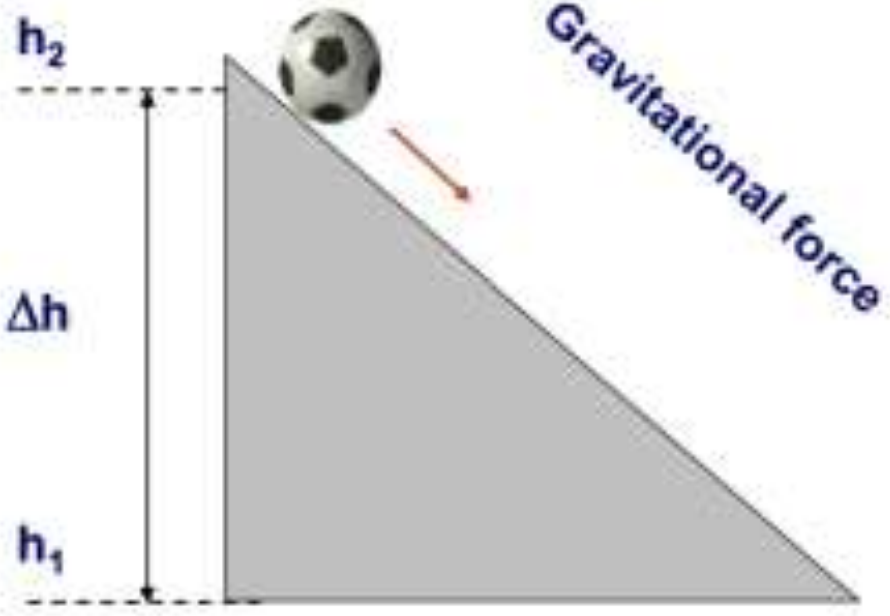


Basic Principles

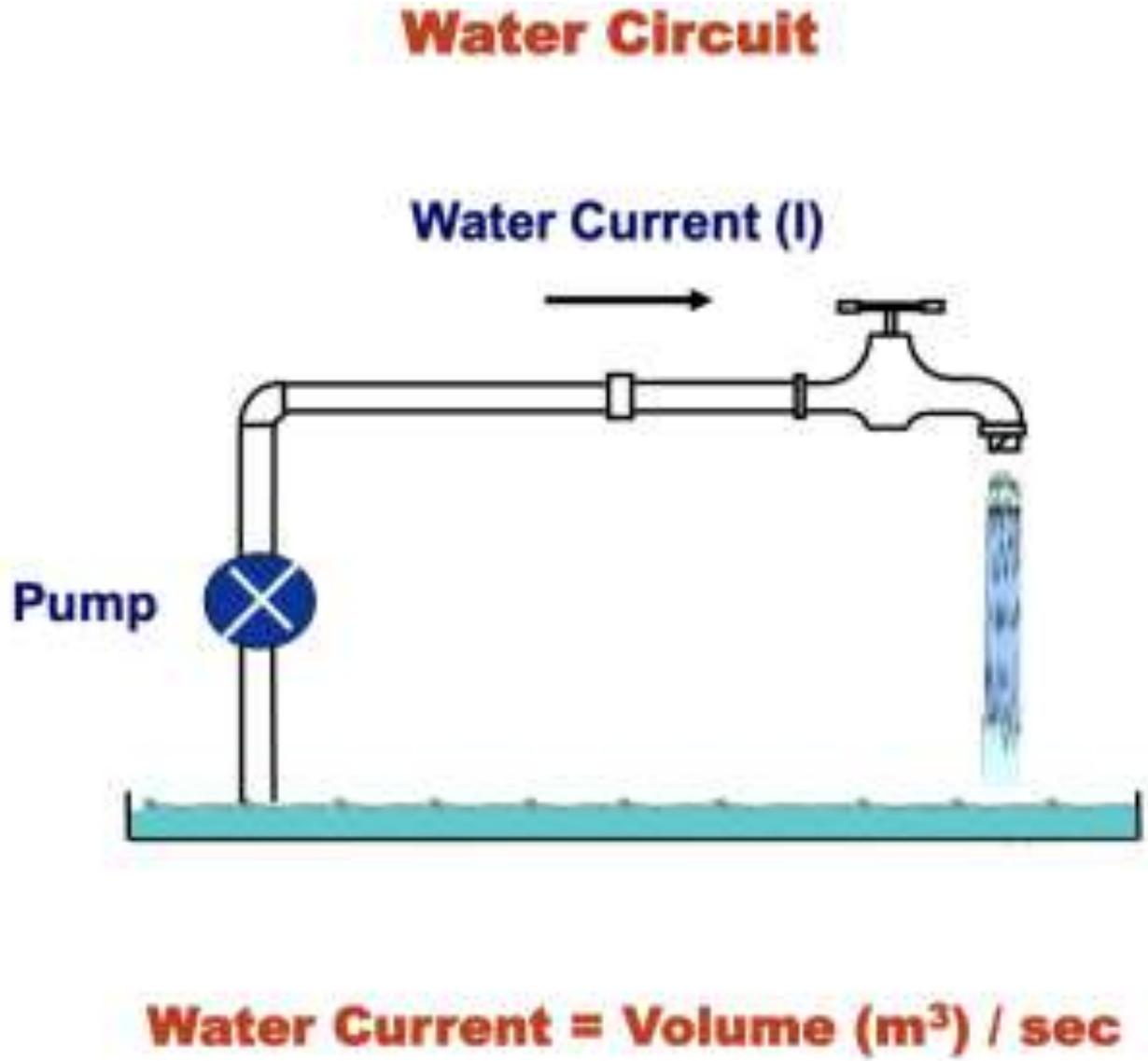
Basic Principle of Circuit

Mechanical Example	Inclined Surface

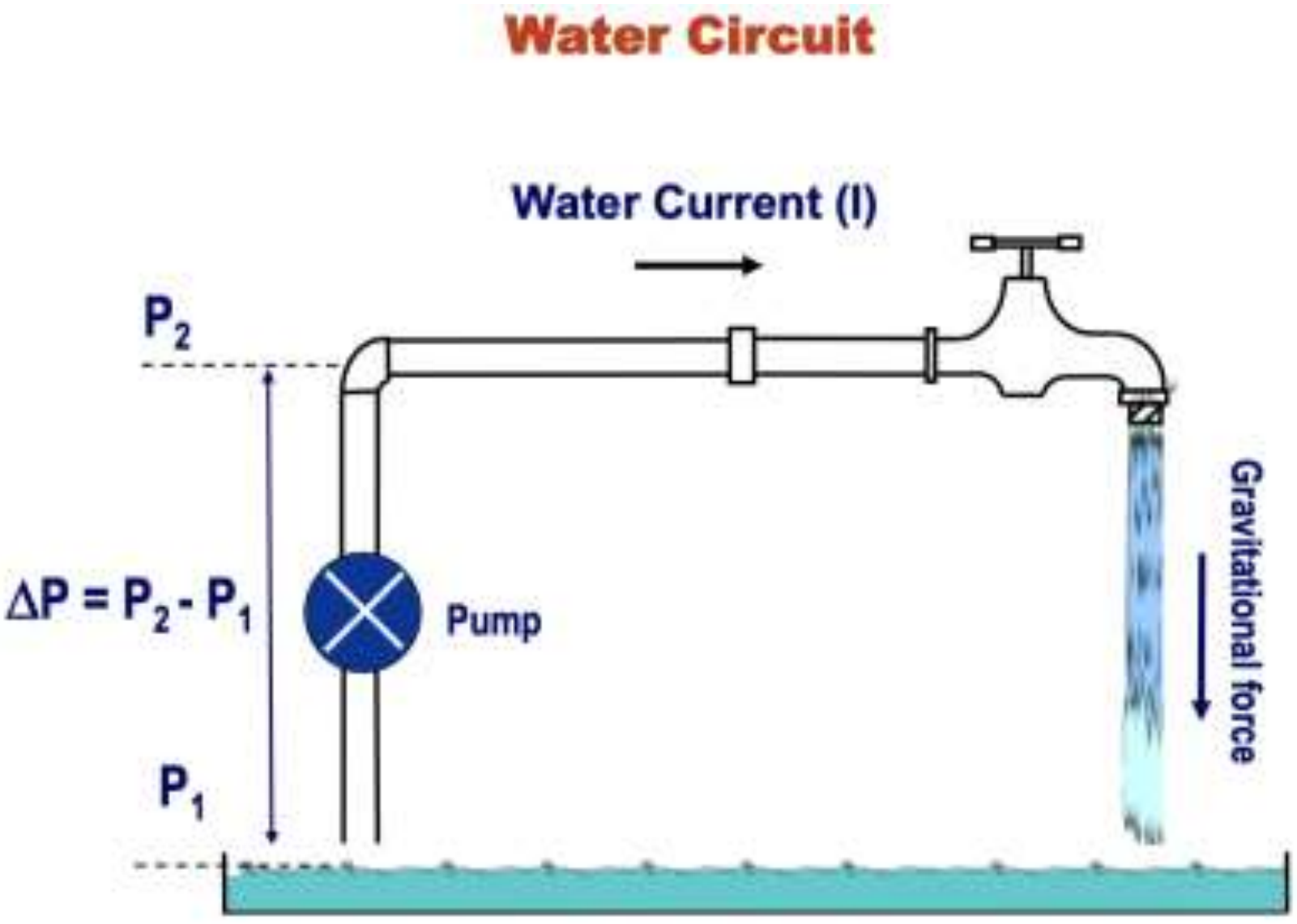




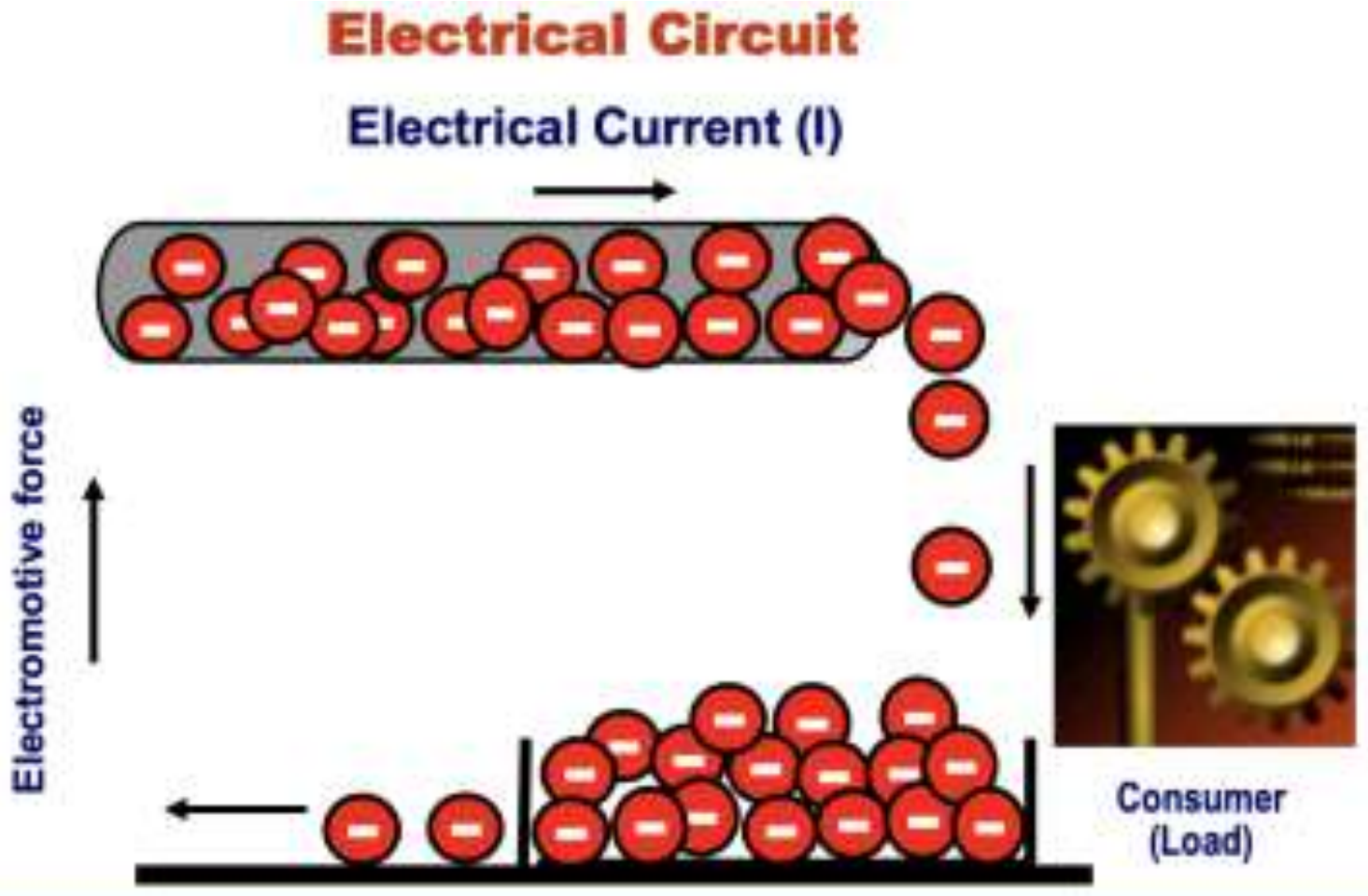
Basic Principles



Basic Principles

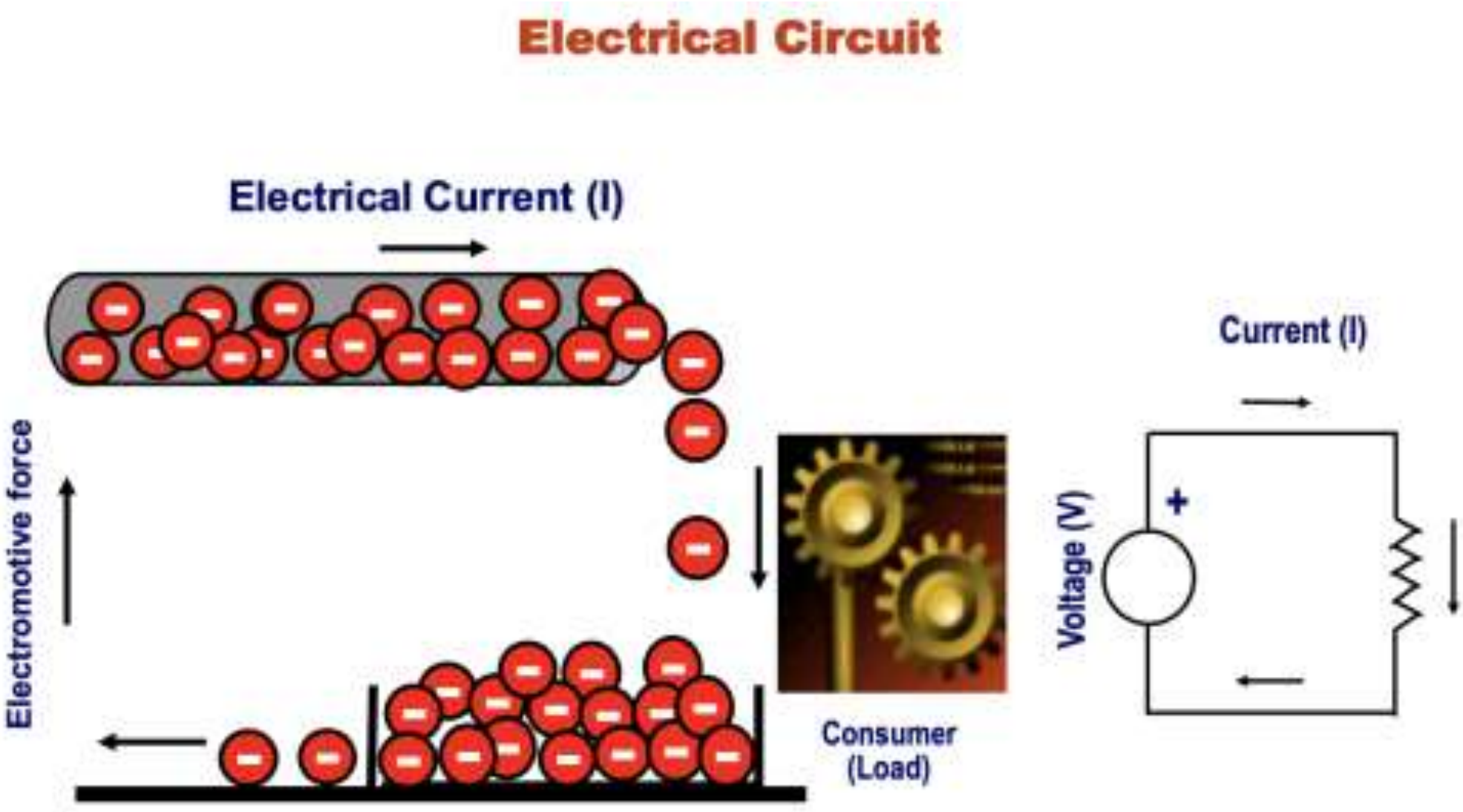


Basic Principles

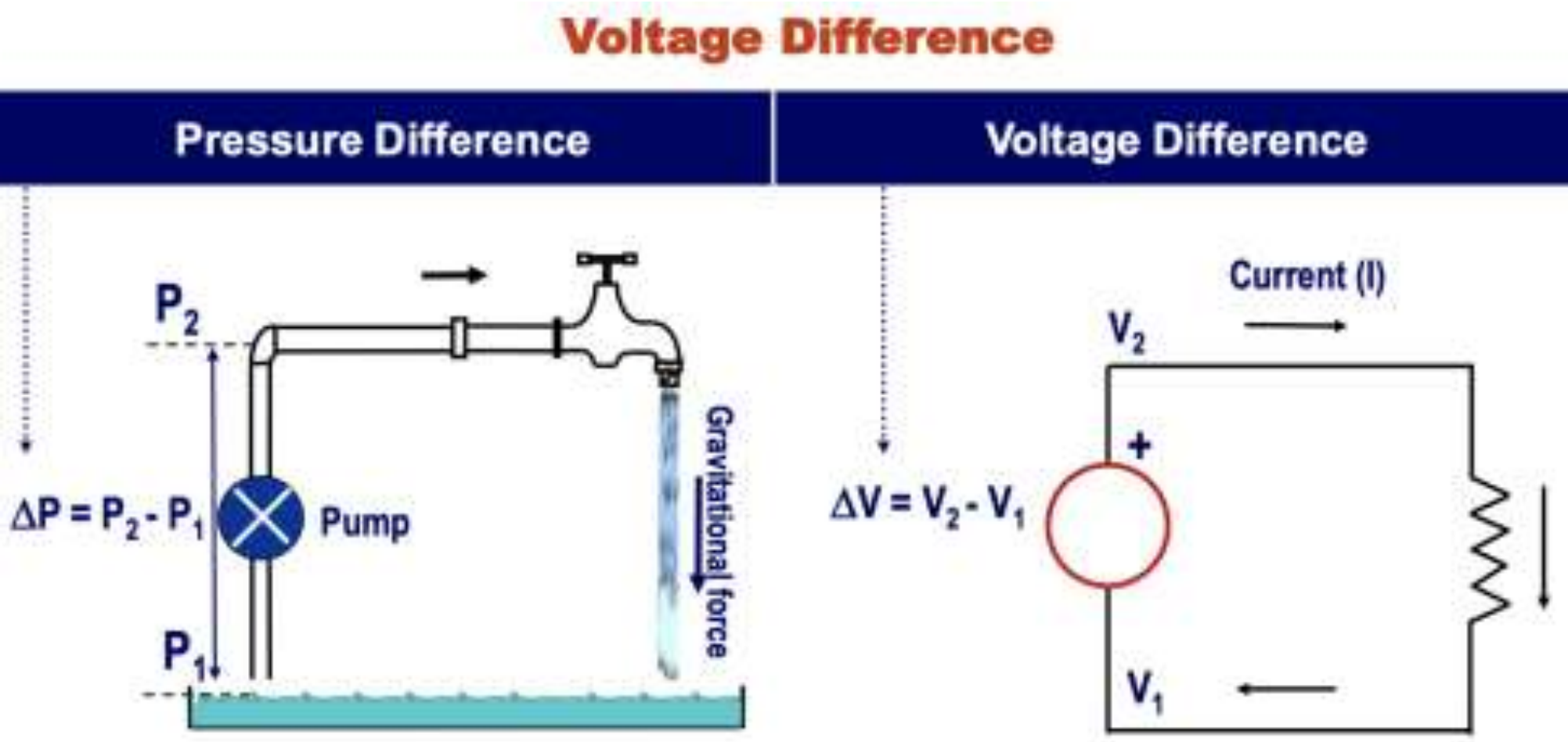


Electrical Current = No. of electrons / sec
= 1 Coulomb / sec
6.3×10^{18} electrons / sec = 1 Amper

Basic Principles



Basic Principles



Basic Principles

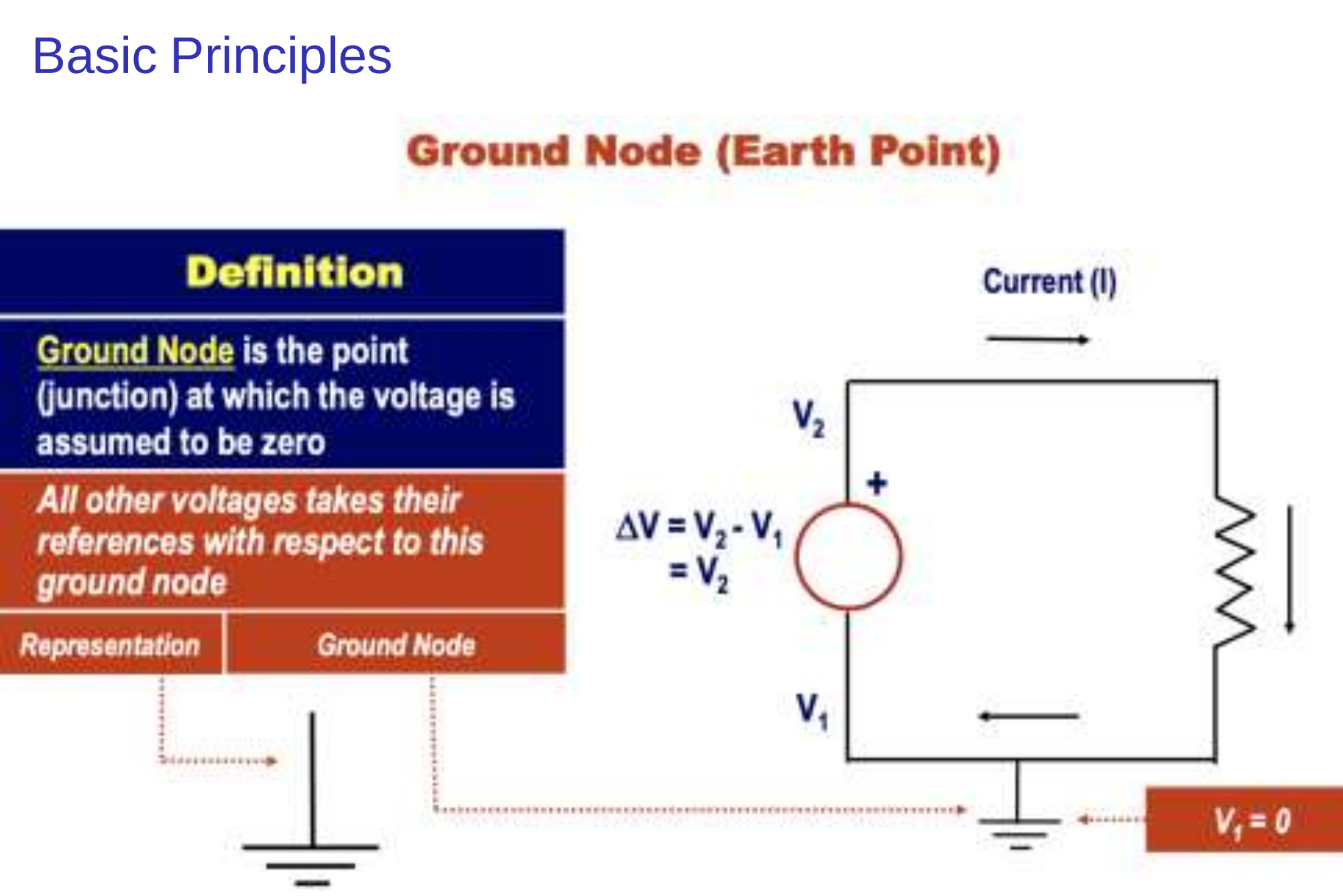
Ground Node (Earth Point)

Definition

Ground Node is the point (junction) at which the voltage is assumed to be zero

All other voltages takes their references with respect to this ground node

Representation	Ground Node



Basic Principles

Ground Node (Earth Point)

Definition

Ground Node is the point (junction) at which the voltage is assumed to be zero

All other voltages takes their references with respect to this ground node



Ground Node (Black Terminal)

Measured Node (Red Terminal)



Basic Principles

Electrical Current

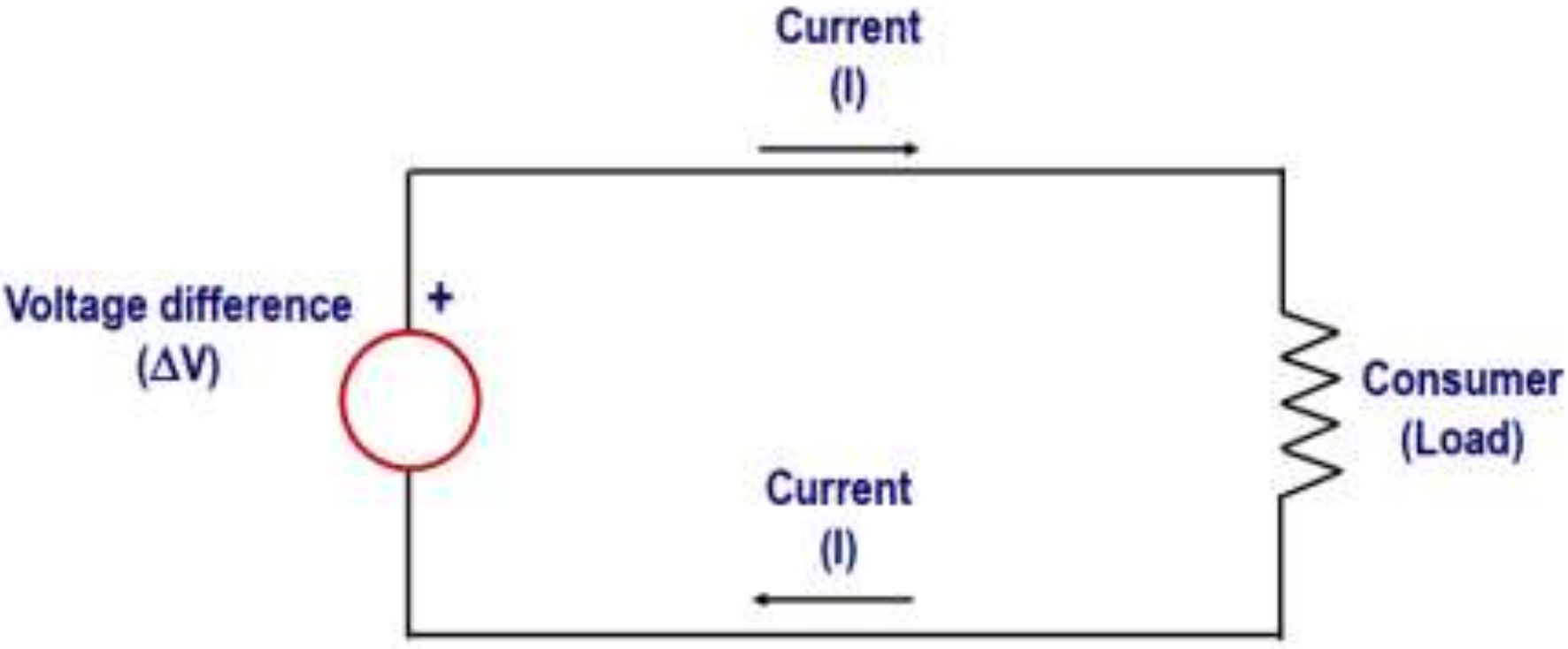
Current = no. of electrons transferred / time duration

$I = \Delta Q / \Delta t$

1 Amp = 1 Coulomb / 1 Seconds

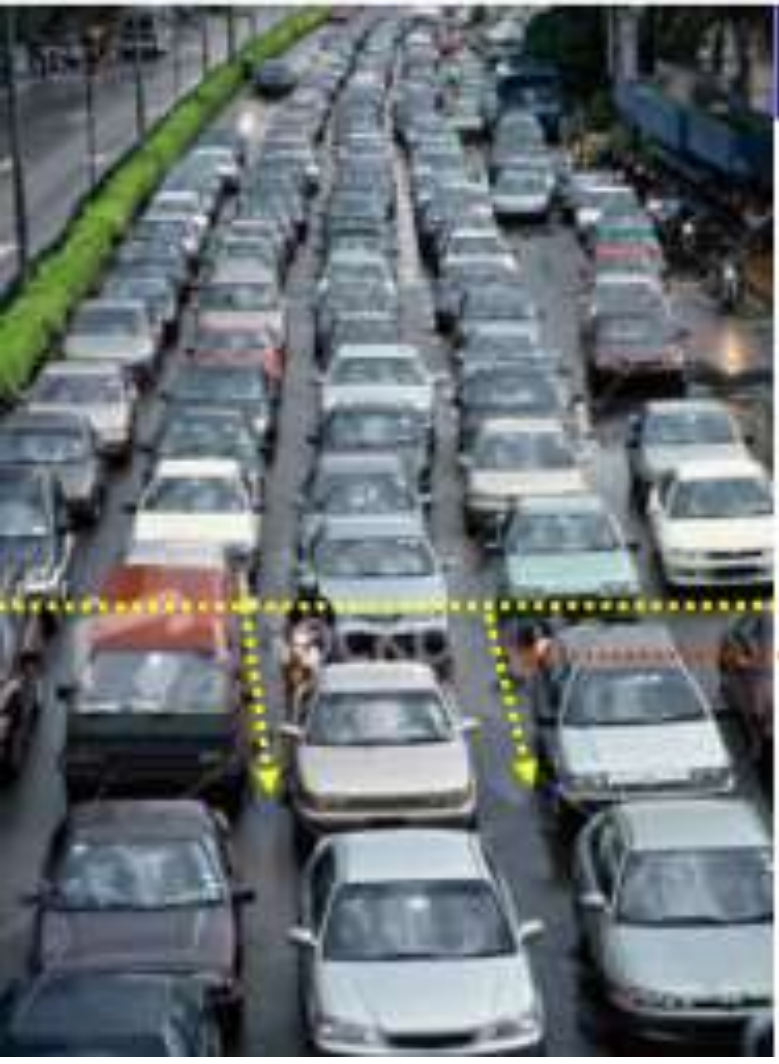
Charge = Current x Time duration

$\Delta Q = I \times \Delta t$



Basic Principles

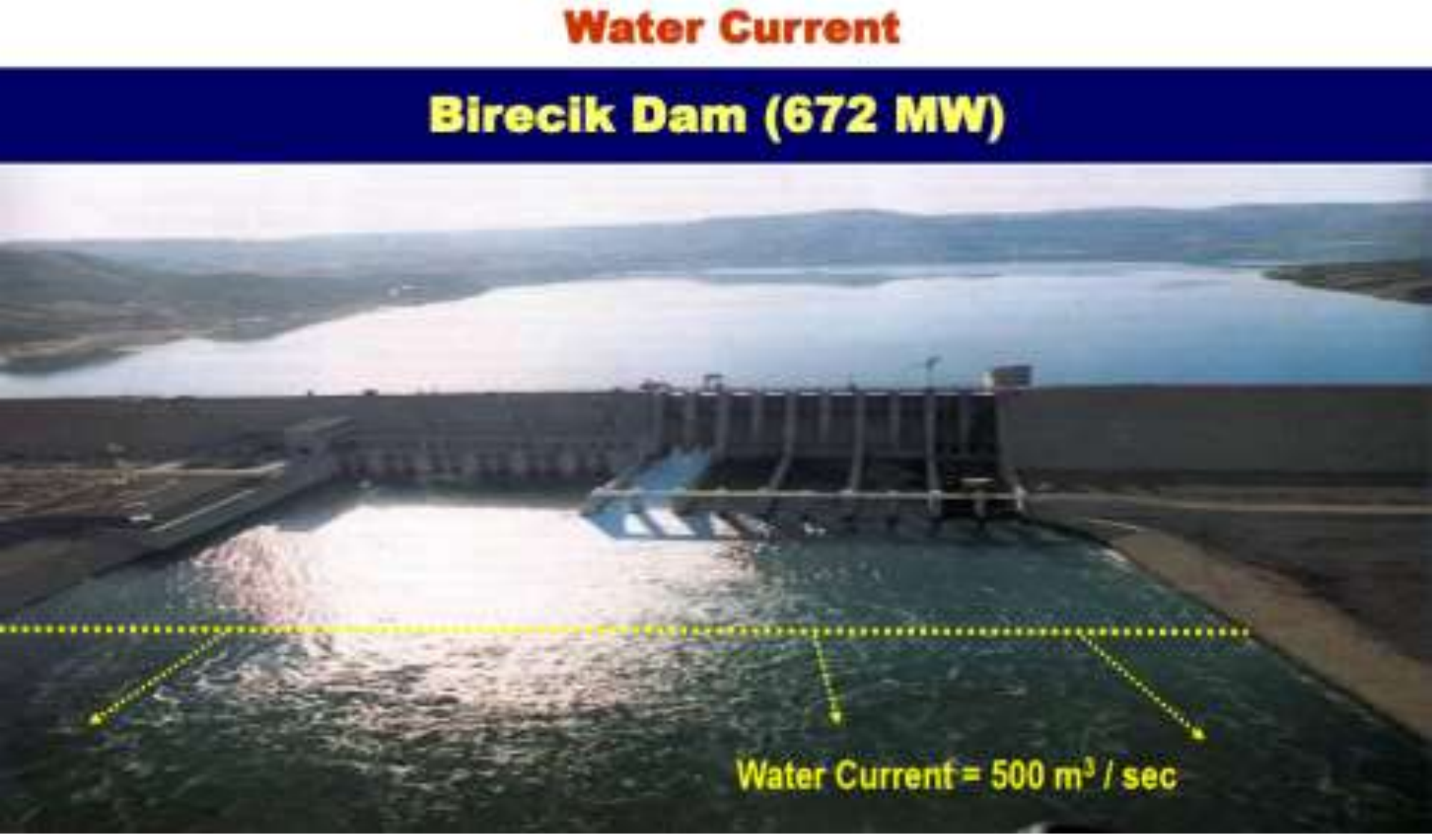
Traffic Current



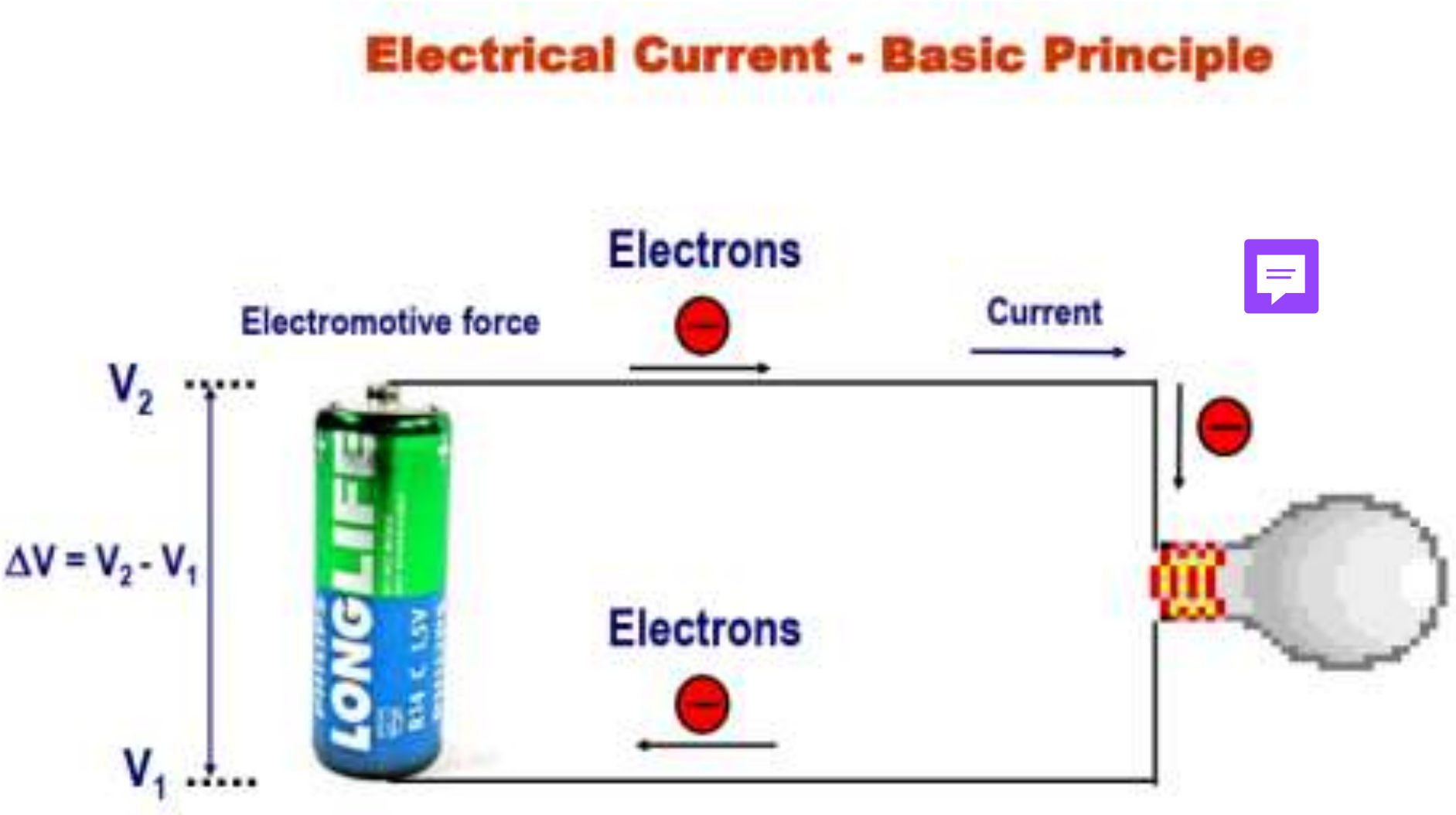
Cars Flowing in a Highway

Traffic Current = Cars / minute

Basic Principles



Basic Principles



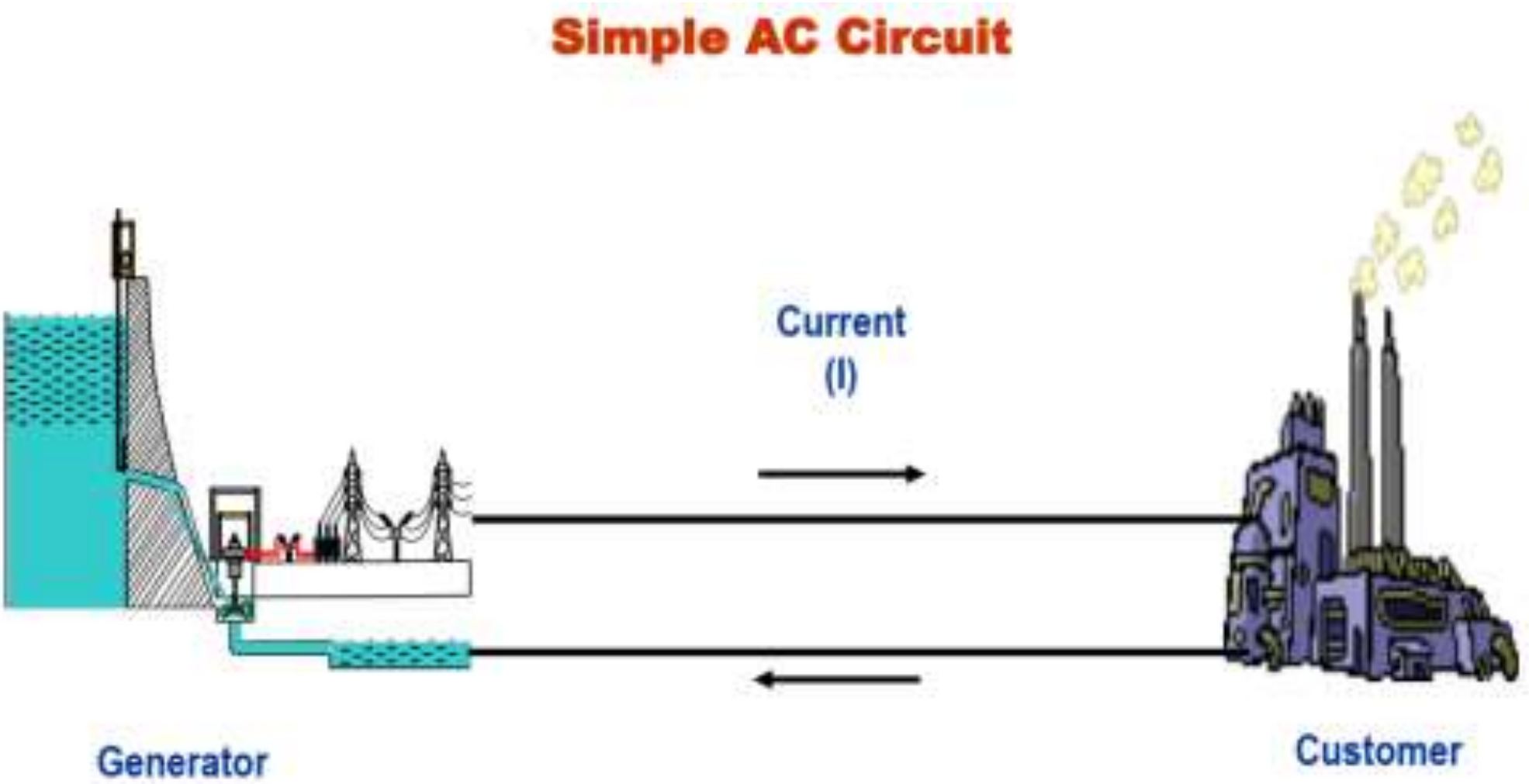
Basic Principles

Electrical Current
DC (Direct Current) Sources





Basic Principles



Basic Principles

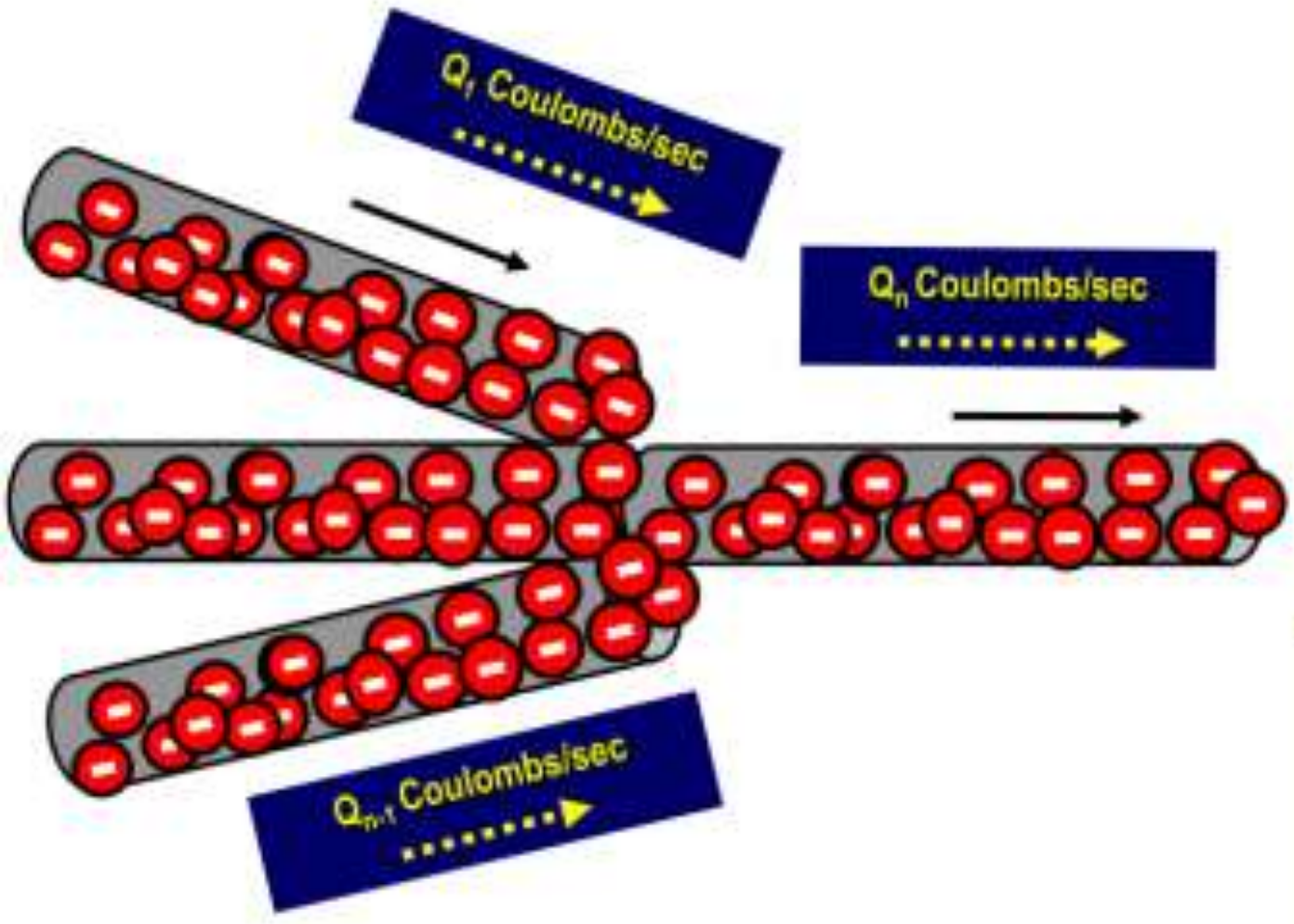
Kirchoff's Current Law (KCL)

Basic Principle	Balance
Σ Cars entering = Σ Cars leaving	Cars entering: 370 Cars leaving: 120 + 65 + 55 + 130 = 370



Basic Principles

Kirchoff's Current Law (KCL)



Charges entering

Q_1

Q_2

Q_{n-1}

\+

Q_{in}

Charges leaving

$Q_{out} = Q_n$

Balance

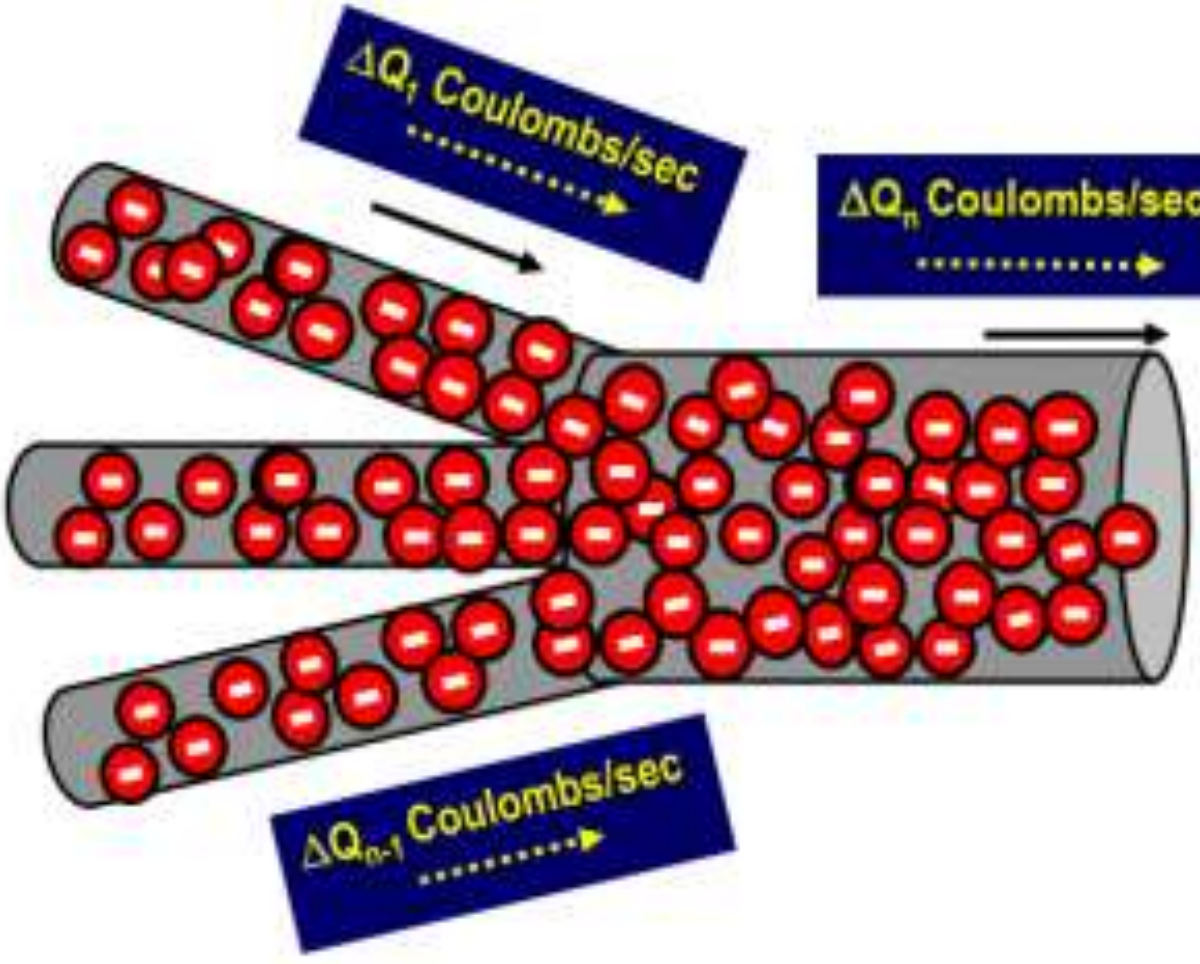
$Q_{in} = Q_{out}$ or

$Q_{in} - Q_{out} = 0$ or

$\Sigma Q = 0$

Basic Principles

Kirchoff's Current Law (KCL)



or

$\Sigma \Delta Q = 0$

or

$\Sigma \Delta Q / \Delta t = 0$

or

$\sum_{i=1}^{i=n} I_i = 0$

Kirchoff's First Law
or
Kirchoffs Current Law

Basic Principles

Mechanical Force

Definition

$F = m \times a$

Force needed to accelarate 1 kg of mass to 1 meter / sec^2 is defined as 1 Newton

1 Newton = 1 kg x 1 meter / sec^2
1000 Newton = 1000 kg x 1 meter / sec^2



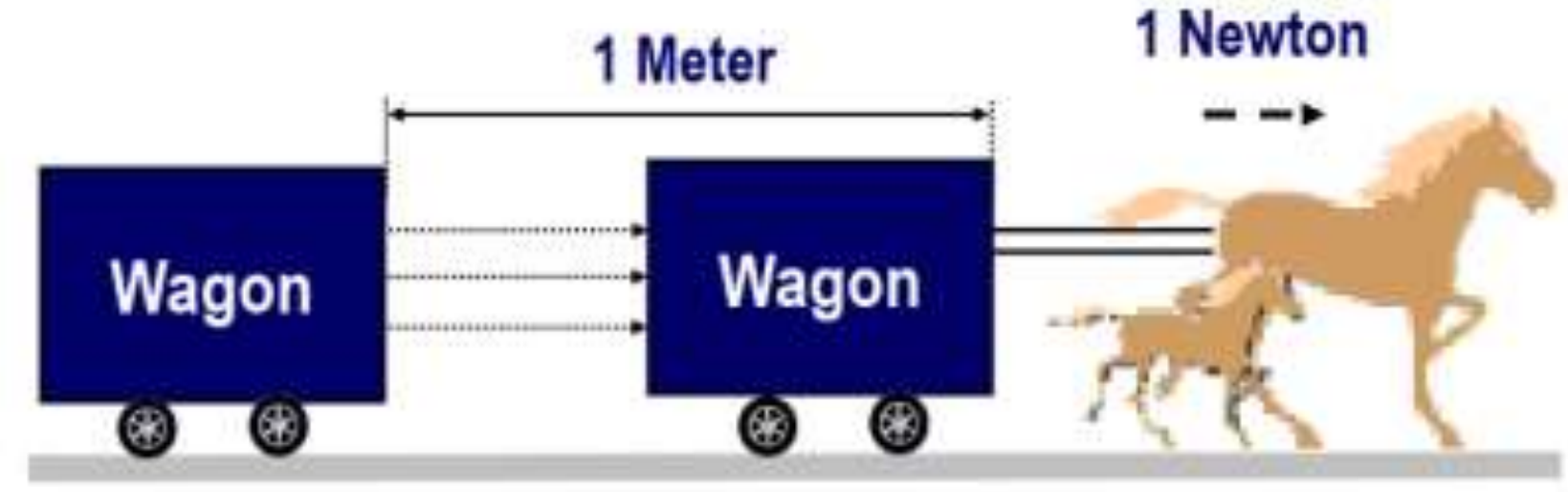
Basic Principles

Mechanical Energy

Definition

1 Joule is the energy needed to move a mass 1 meter by using 1 Newton force

1 Joule = 1 Newton x 1 Meter



Basic Principles

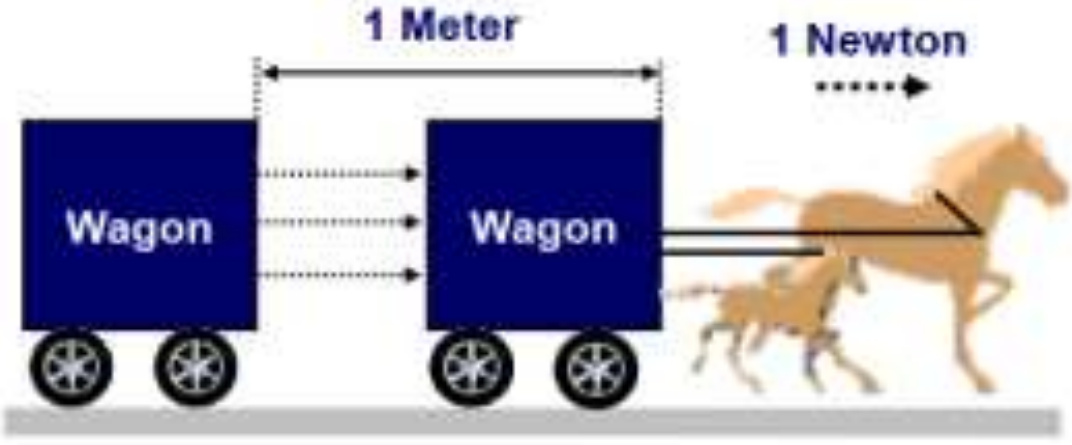
Mechanical Energy vs Electrical Energy

Equivalance

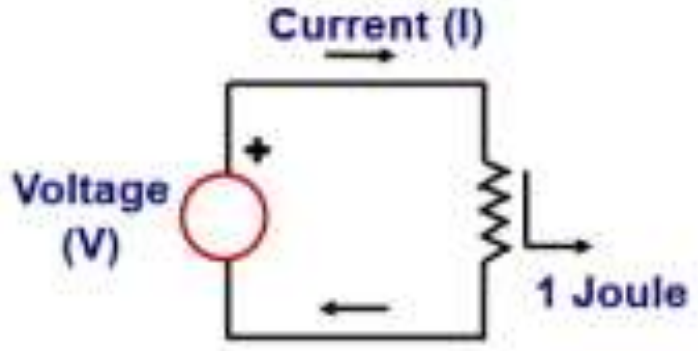
Mechanical Energy = Electrical Energy

Mechanical Work = Electrical Work

The same amount of energy may be spent out by using electricity



Mechanical Energy (Work) = 1 Joule



Electrical Energy (Work) = 1 Joule

Basic Principles

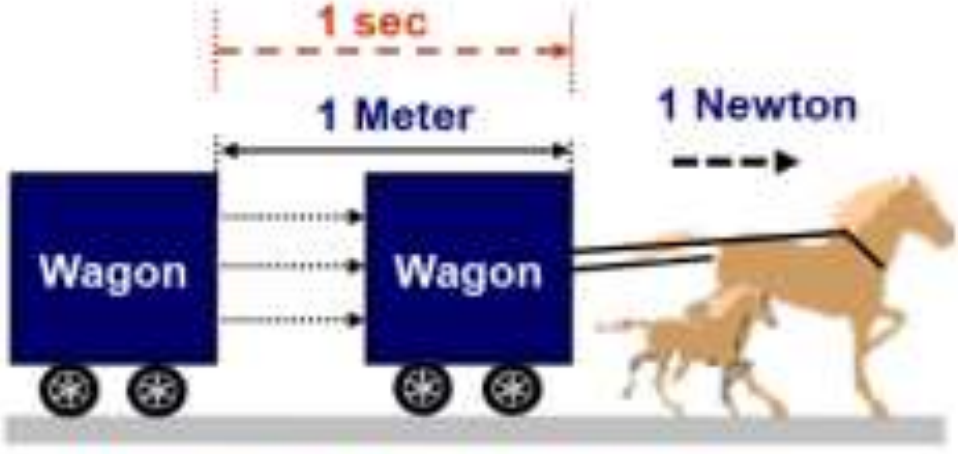
Electrical Power

Definition

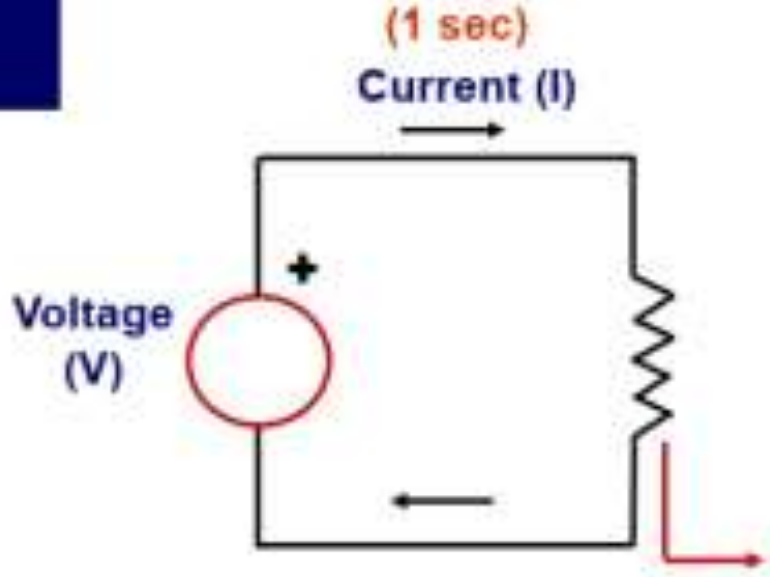
Similar to mechanical power, electrical power is the work done within a certain unit of time, i.e. one second or one hour

Elecrical Power = Electrical Energy / Duration

= 1 Joule / sec



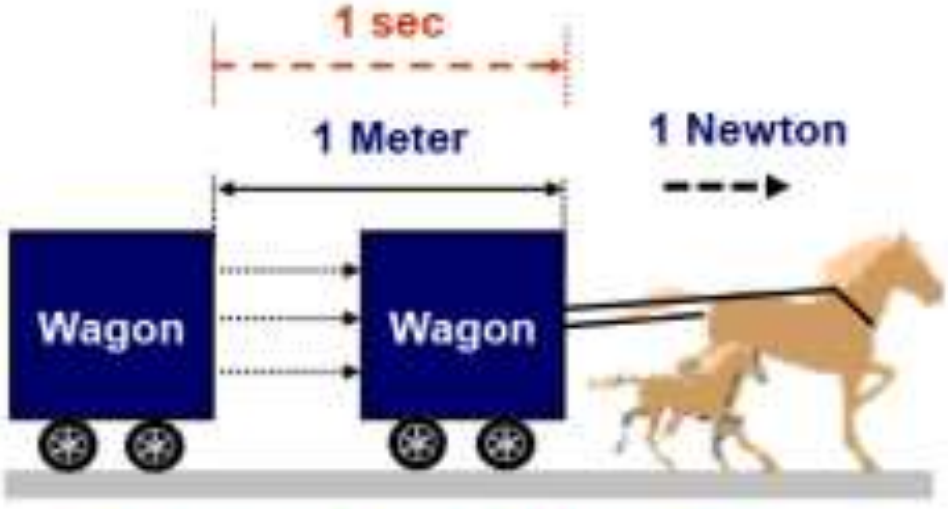
Mechanical Power = 1 Joule / sec.



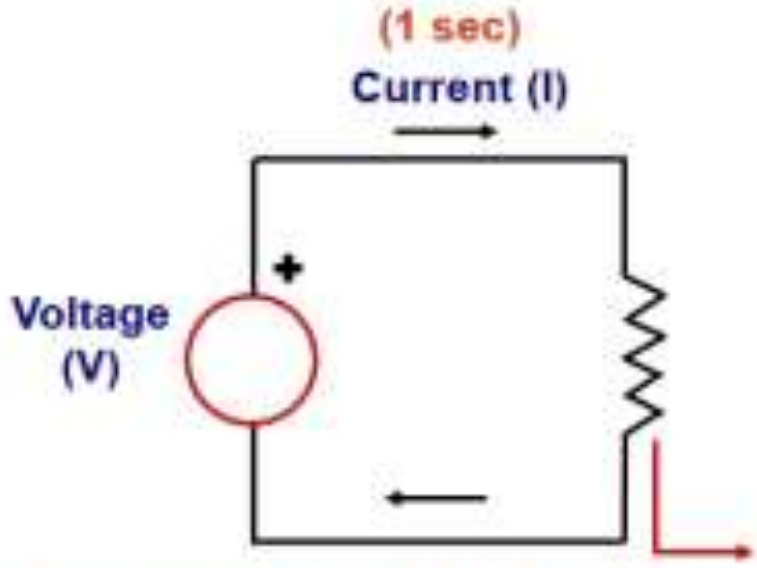
Electrical Power = 1 Joule / sec.

Basic Principles

Equivalance | *Mechanical Power = Electrical Power*



Mechanical Power = 1 Joule / sec.

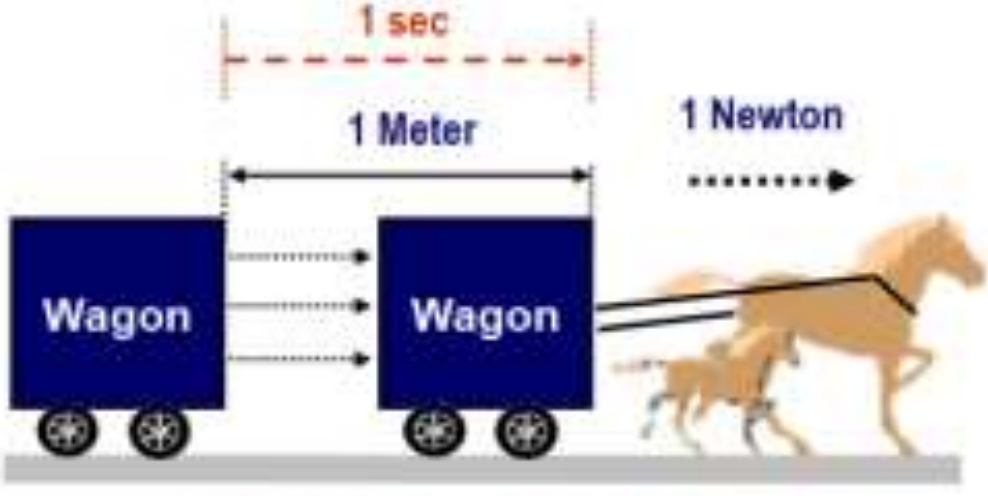


Electrical Power = 1 Joule / sec.

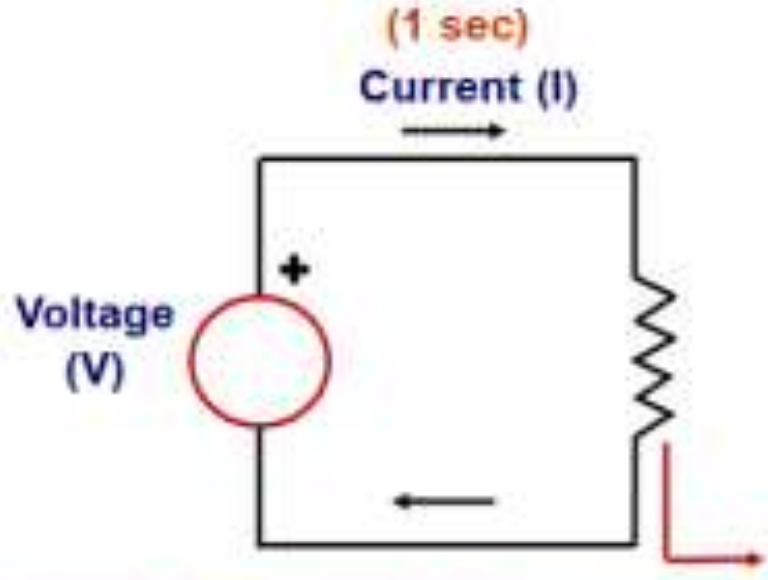
Basic Principles

Electrical Power

Definition	1 Joule = 1 Watt x second
1 Joule / second = 1 Watt (1 Joule energy is spent within 1 second)	1 Horse Power = 746 Watts = 0.746 kWatt



1 Joule / sec = 1 Watt

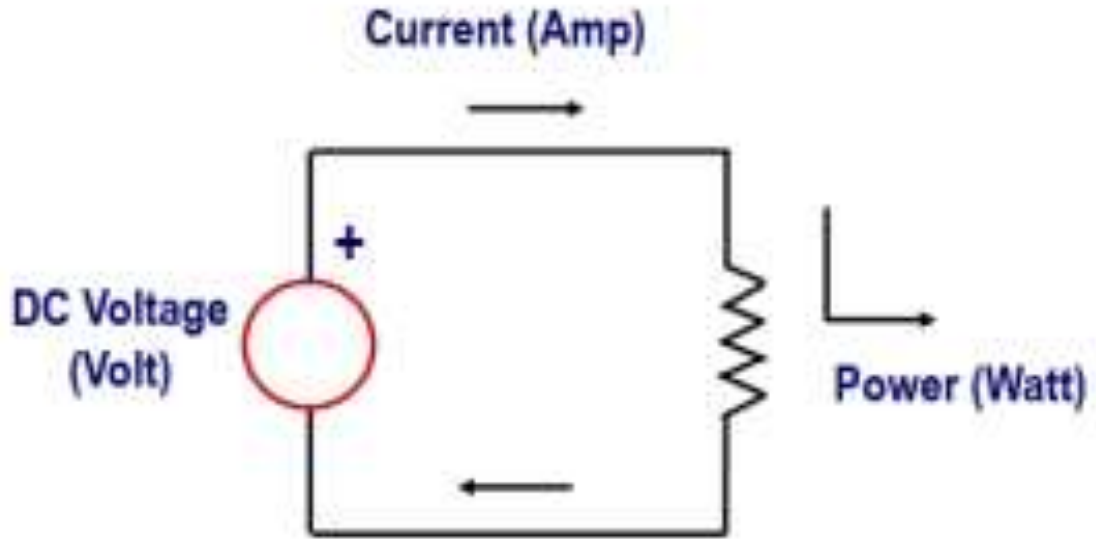


Electrical Power = 1 Joule / sec. = 1 Watt

Basic Principles

Electrical Power

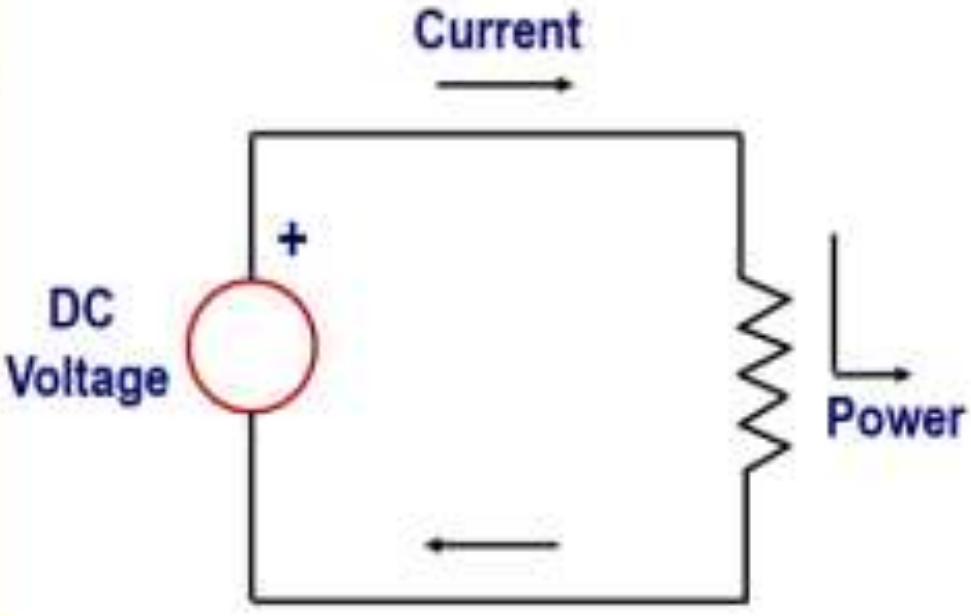
Definition	Power = Voltage x Current
P = V x I (Watt) = (Volt) x (Amp)	



Basic Principles

Voltage

Definition
Power = Voltage x Current *or* *P = V x I*
Voltage = Power / Current *or* *V = P / I*



Basic Principles

Voltage

Definition

Power = Voltage x Current

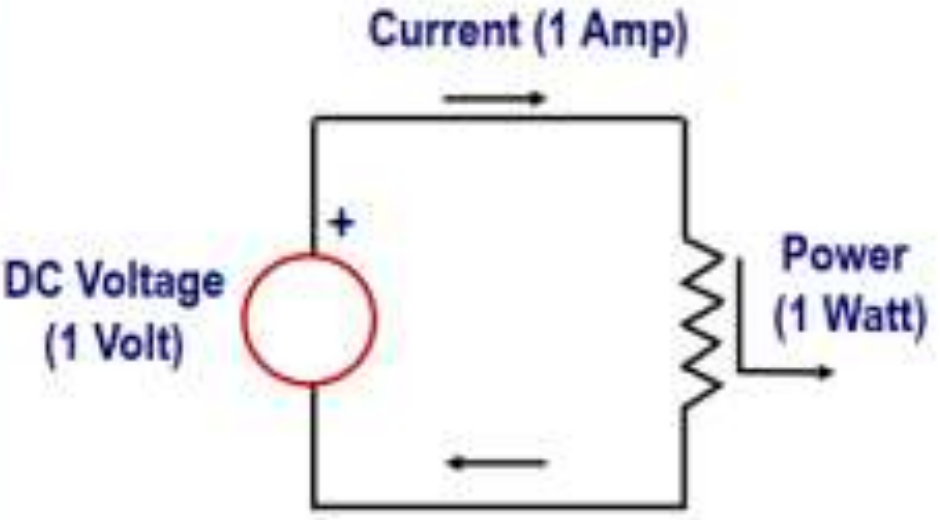
or

Voltage = Power / Current

or

V = P / I

1 Volt = 1 Watt / 1 Amp

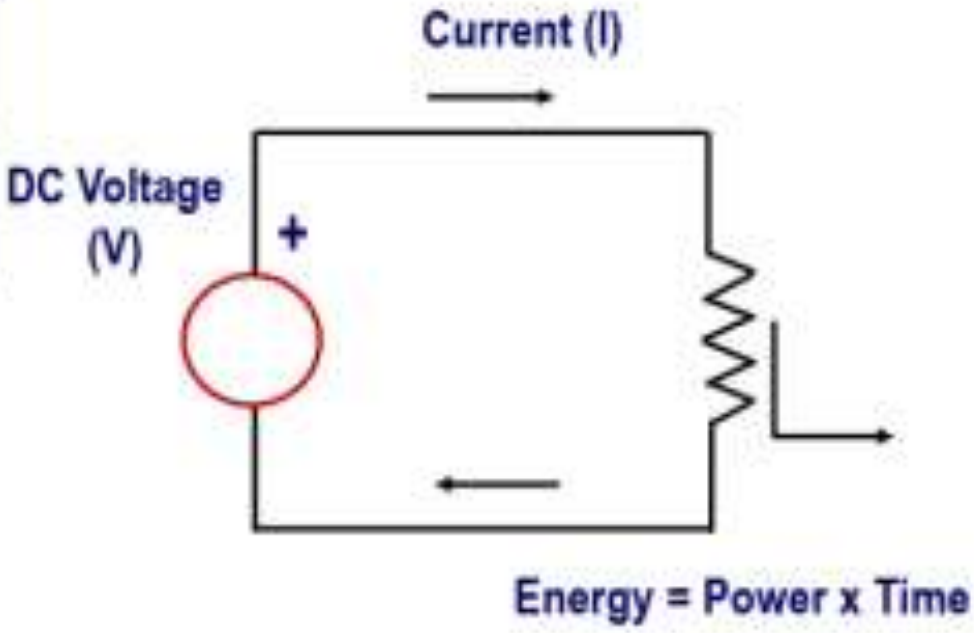


Basic Principles

Electrical Energy

Definition

Energy = Power x Time
(Watt-sec) (Watt) (second)



Basic Principles

Unit of Electrical Energy

Definition

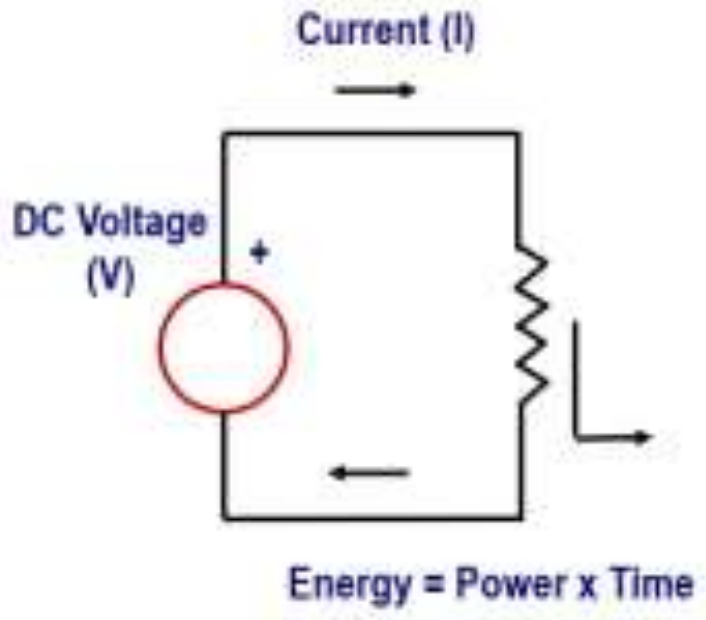
Energy = Power x Time
(Watt-sec) (Watt) (second)

Energy = Power x Time
(KiloWatt-hour) (KiloWatt) (hour)

x 1000 *x 1000* *x 3600*

1 KiloWatt - hour = 1000 x 3600 Watt x seconds
= 3 600 000 Joules

1 KiloWatt = 1000 Watts
1 Hour = 3600 seconds



Basic Principles

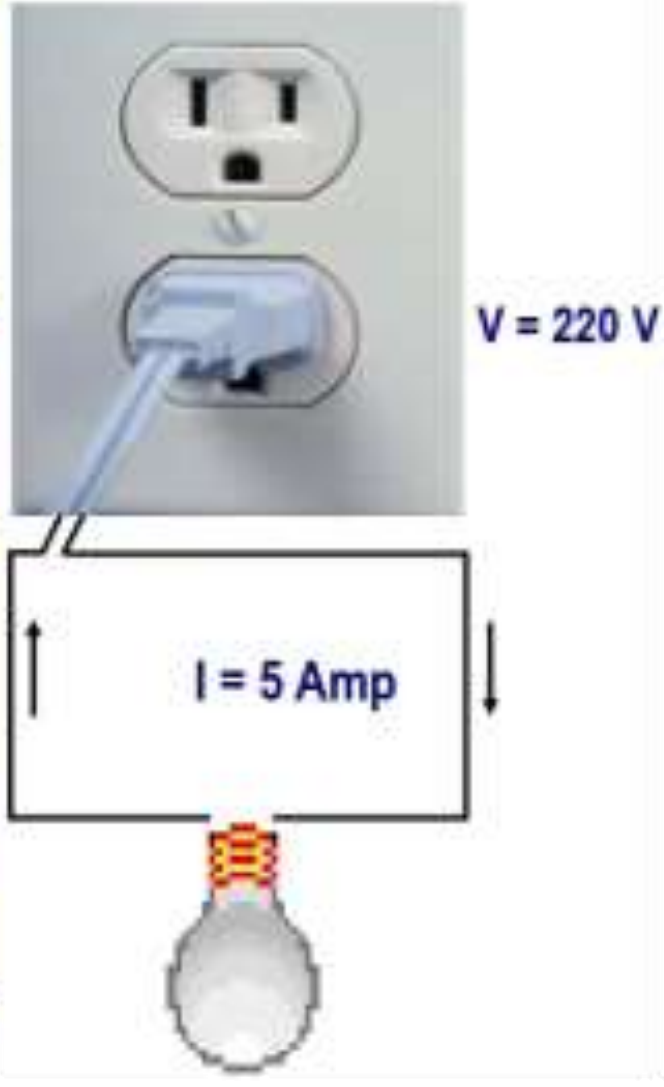
Electrical Energy

Example

Calculate the monthly payment for the energy consumed by the lamp shown

Source voltage is 220 Volt
Current drawn by the lamp is 5 Amp
Price of electrical energy is 12 Cents / kWh

Power = Voltage x Current
P = V x I
P = 220 x 5 = 1100 Watts
Energy = P x Δt
= 1100 Watts x (24 hours /day x 30 days/month)
= 792000 Watt hours = 790.2 kWh
Monthly payment = 790.2 x 12 Cents / month



Basic Principles

Alternative Definition of Voltage

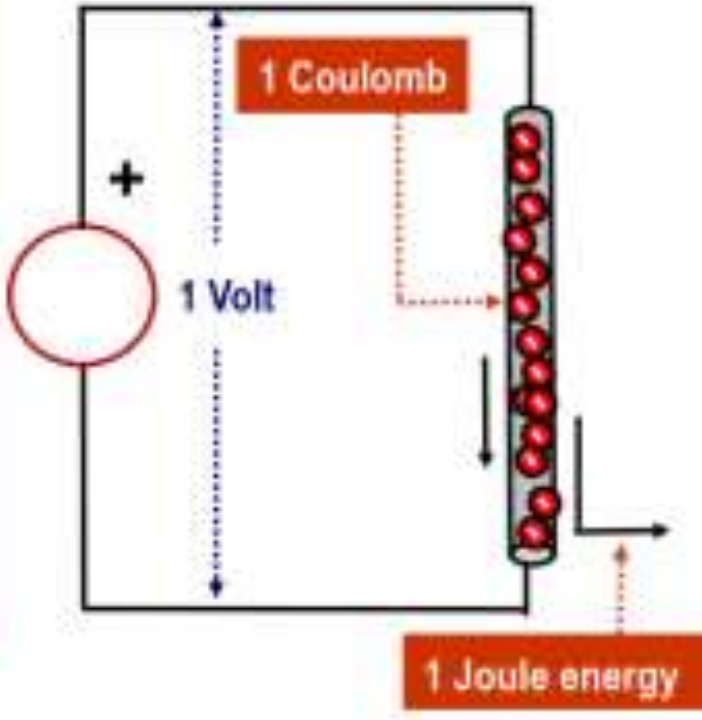
1 Volt = *1 Watt / 1 Amp*
= *(1 Joule /sec) / 1 Amp*
= *1 Joule / (1 Amp x sec)*
= *1 Joule / 1 Coulomb* (*)

(*) *Remember that 1 Amp = 1 Coulomb / 1 sec*

1 Volt is the voltage needed;

- to move 1 Coulomb of electrical charge,
- to spend 1 Joule of energy for this movement

in a conductor



Chapter 1 – Circuit Variables (4 of 21)

Section 1.4

Voltage

Definition – the energy per unit charge created when +

and – charges are separated.

Defining equation: $$v = \frac{dw}{dq}$$

where

v is voltage in volts [V]

w is energy in joules [J]

q is charge in coulombs [C]

Chapter 1 – Circuit Variables (5 of 21)

Section 1.4 (continued)

Current

Definition – the time rate of change of charge flow.

Defining equation: $$i = \frac{dq}{dt}$$

where

i is current in amperes [A]

q is charge in coulombs [C]

t is time in seconds [s]

Chapter 1 – Circuit Variables (6 of 21)

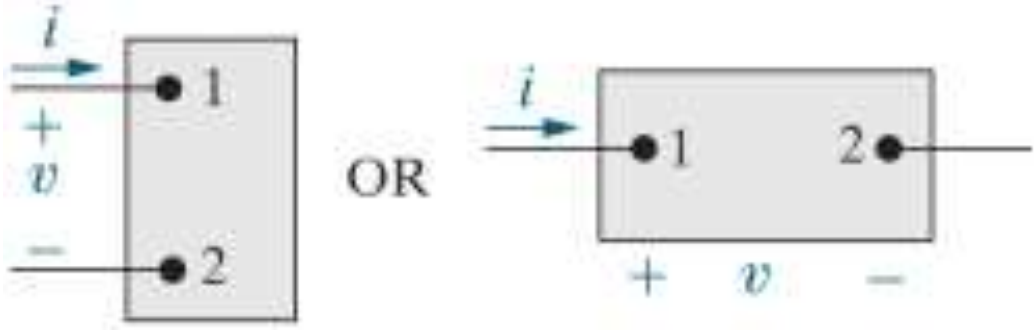
Section 1.5

Ideal Basic Circuit Element

Ideal – the element can be described solely in terms of the relationship between its voltage and its current.

Basic – the element cannot be sub-divided into simpler elements.

Circuit Element – the element has two terminals used to connect it to other elements and form a circuit.



Chapter 1 – Circuit Variables (7 of 21)

Section 1.6

Power

Definition – the time rate of change of energy.

Defining equation: $$p = \frac{dw}{dt}$$

where

p is power in watts [W]

w is energy in joules [J]

t is time in seconds [s]

Note that energy is the integral over time of power!

Chapter 1 – Circuit Variables (8 of 21)

Section 1.6 (continued)

More on power:

$$p = \frac{dw}{dt} = \frac{dw}{dq} \cdot \frac{dq}{dt} = vi$$

$\boxed{p = vi}$ Memorize this!!

Chapter 1 – Circuit Variables (9 of 21)

Section 1.6 (continued)

More on power:

Actually, $\boxed{p = \pm vi}$

How do you know which sign to use???

The passive sign convention:

"Whenever the reference direction for the current in an element is in the direction of the reference voltage drop across the element, use a positive sign in any expression that relates the voltage to the current. Otherwise, use a negative sign."

Chapter 1 – Circuit Variables (10 of 21)

Section 1.6 (continued)

The practical passive sign convention:



If the current arrow points to the + terminal of the voltage, use a + sign in the equation. Otherwise, use a − sign.

For the ideal basic circuit element above,

$$p = +vi = vi$$

Chapter 1 – Circuit Variables (11 of 21)

Section 1.6 (continued)

Passive sign convention examples:



$p = -vi$



$p = -vi$



$p = +vi = vi$

Chapter 1 – Circuit Variables (12 of 21)

Section 1.6 (continued)

What is the meaning of positive power and negative power?

When the power associated with a circuit element is negative, the circuit element is generating (delivering, supplying) power to the circuit.

When the power associated with a circuit element is positive, the circuit element is absorbing (dissipating) power from the circuit.

Chapter 1 – Circuit Variables (13 of 21)

Section 1.6 (continued)

Examples of interpreting the algebraic sign of power:



Is power supplied or absorbed? Supplied



Is power supplied or absorbed? Supplied



Is power supplied or absorbed? Absorbed

Chapter 1 – Circuit Variables (14 of 21)

Chapter 1 examples:

Find the total charge entering terminal 1.



$$i = 0, \qquad t < 0$$

$$i = 20e^{-5000t} \text{ A}, \qquad t \geq 0$$

$$i = \frac{dq}{dt} \qquad \Rightarrow \qquad q_{\text{total}} = \int_0^\infty i\,dt + q(0)$$

$$q_{\text{total}} = \int_0^\infty 20e^{-5000t}\,dt + 0 = \frac{20}{-5000}e^{-5000t}\Big|_0^\infty$$

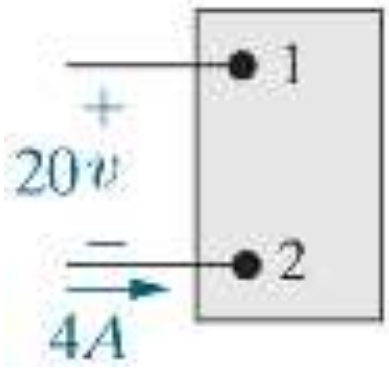
$$= \frac{20}{-5000}(0) - \frac{20}{-5000}(1)$$

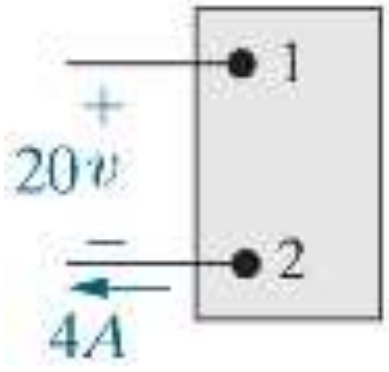
$$= 0.004 \text{ C} = 4\,\text{mC} = 4000\,\mu\text{C}$$

Chapter 1 – Circuit Variables (16 of 21)

Chapter 1 examples: Assessment Problem 1.5 (modified)

Find the power associated with the circuit elements below. State whether the power is generated or absorbed.





$p = -vi = -(20)(4) = -80$ W

$\Rightarrow$ 80 W generated

$p = vi = (20)(4) = 80$ W

$\Rightarrow$ 80 W absorbed

Chapter 1 – Circuit Variables (17 of 21)

Chapter 1 examples: Find the total energy delivered to the circuit element.



$$i = 0, \qquad t < 0$$

$$i = 20e^{-5000t} \text{ A}, \qquad t \geq 0$$

$$v = 0, \qquad t < 0$$

$$v = 10e^{-5000t} \text{ kV}, \qquad t \geq 0$$

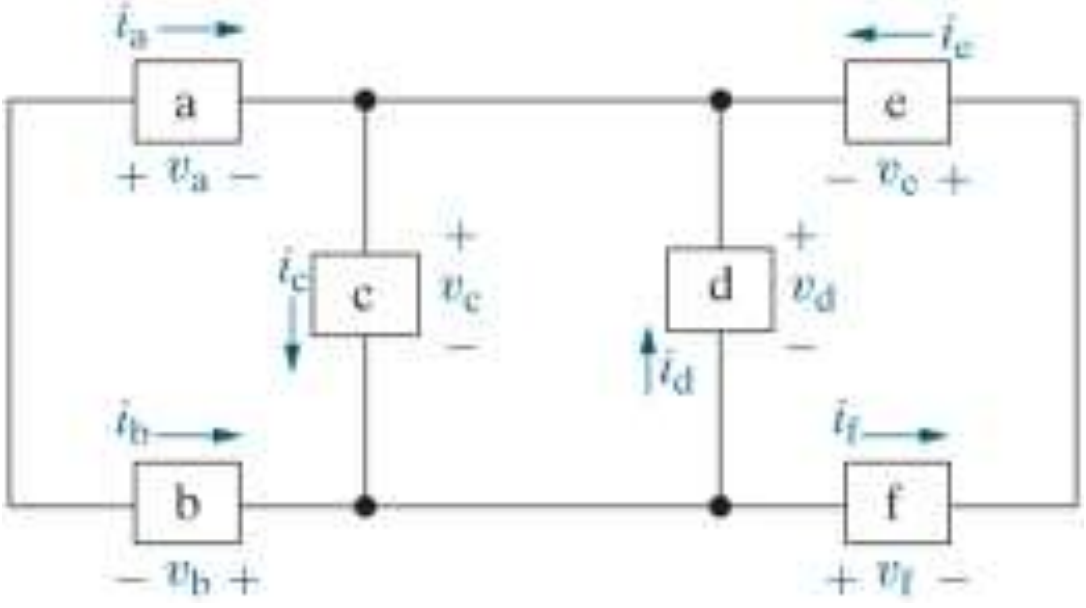
$$p = vi = \frac{dw}{dt} \qquad \Rightarrow \qquad w = \int_0^\infty vi \; dt + w(0)$$

$$w = \int_0^\infty (20e^{-5000t})(10^4 e^{-5000t})dt + 0 = 2 \times 10^5 \int_0^\infty e^{-10{,}000t} dt$$

$$= \frac{2 \times 10^5}{-10{,}000} e^{-10{,}000t} \Big|_0^\infty = \frac{2 \times 10^5}{-10{,}000}(0 - 1) = 20 \text{ J}$$

Chapter 1 – Circuit Variables (19 of 21)

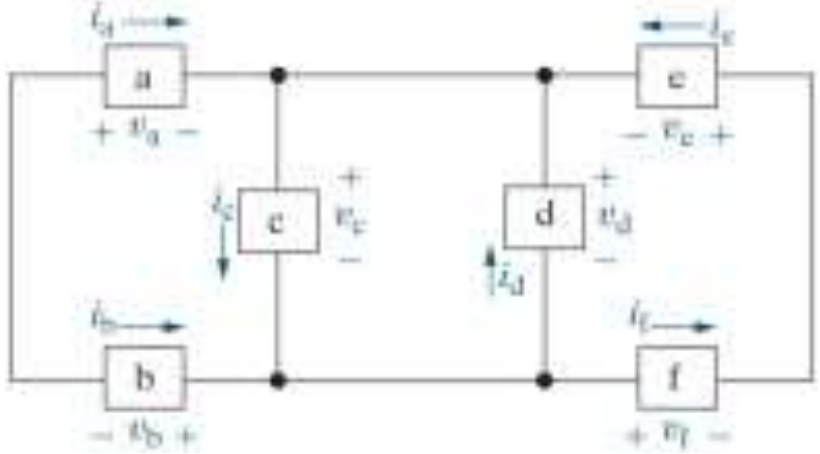
Chapter 1 examples: Find the power associated with each circuit element in the circuit below.



$v_a = -8\ \text{V}$ $v_b = -2\ \text{V}$ $v_c = 10\,\text{V}$ $v_d = 10\,\text{V}$ $v_e = -6\,\text{V}$ $v_f = -4\text{V}$

$i_a = 7\ \text{A}$ $i_b = -7\ \text{A}$ $i_c = 15\ \text{A}$ $i_d = 5\ \text{A}$ $i_e = 3\ \text{A}$ $i_f = 3\ \text{A}$

Chapter 1 – Circuit Variables (20 of 21)



Comp.	v[V]	i[A]	sign	Eqn.	p[W]	sup/abs?
a	–8	7	+vi	(–8)(7)	–56	supply
b	–2	–7	–vi	–(–2)(–7)	–14	supply
c	10	15	+vi	(10)(15)	150	absorb
d	10	5	–vi	–(10)(5)	–50	supply
e	–6	3	+vi	(–6)(3)	–18	supply
f	–4	3	+vi	(–4)(3)	–12	supply
				Total	0!!!	